INCH-POUND
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DETAIL SPECIFICATION

PROJECTILE, 105 MILLIMETER, HE, RA- M913/M927, DELAY ASSEMBLY FOR

Inactive for new design after 21 August 1998

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication and packaging of parts and assembly of the delay assembly for the 105MM, High Explosive, Rocket Assisted, M913 / M927 Projectiles.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents sited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-217 - Potassium Perchlorate (for pyrotechnics)
MIL-T-48140 - Tungsten Powder (for pyrotechnics)
MIL-V-50433 - Vinyl Alcohol – Acetate Resin Solution (VAAR) (For Use Ammunition)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition lot numbering and ammunition data cards
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing
MIL-STD-1916 - DoD preferred methods for acceptance of product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation

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US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

12926809 - Delay Assembly
13023959 - Radiographic Inspection Method for Artillery Projectiles and Components

(Copies of these drawings may be requested online at Drawing.Request@conus.army.mil or from U.S. Army ARDEC, ATTN: RDAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM B430 - Standard Test Method for Particle Size Distribution of Refractory Metal Powders and Related Compounds by Turbidimetry

(Copies of ASTM standards are available online at http://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 12926809.

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3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Pyrotechnic composition, total volatiles. The total volatiles of the flash, delay and igniter composition at the time of loading shall not exceed 0.10 percent. The flash, delay and igniter compositions shall comply with dwg. 12926809.

3.5 Crimp strength. The washer crimp shall be capable of withstanding a 500 pound minimum load.

3.6 Pyrotechnic porosity test. The pressure recorded at the exit end of the delay assembly; 80 seconds after application of the nitrogen gas shall not exceed 9 PSIG above the zero reference. The pressure recorded at the exit end of the delay assembly 110 seconds after application of the nitrogen gas shall be no greater than 19 PSIG above the zero reference.

3.7 Radiographic inspection. The delay column shall not be missing or be short in length. There shall be no voids or low density areas in the area of individual increments in the delay column. There shall be no evidence of aluminum slivers in the delay column. The flash or igniter composition shall not be missing. The boron potassium nitrate pellets shall not be missing, contain voids greater than 1/64th square inch, or have a gap between the top of the closure greater than 1/8th inch. The delay washer shall not be missing.

3.8 Performance. (See 6.22 for definitions).

3.8.1 Pilot composition. The pilot composition shall be acceptable when the ballistic delay burn times of the pilot composition batches show that the required burn time of 15.25 $\pm$ .75 seconds at 70 degrees F is met by at least one batch or is bracketed by at least one batch on the high and low ends so that an optimal burn time fuel ratio may be determined. The standard deviation of the batch(es) meeting the requirement shall not exceed .62 seconds when ballistically tested at 70 degrees F in accordance with 4.4.7. When ballistically tested at -50 degrees F, the delay shall function to the extent that it ignites the rocket motor. During static spin testing, the samples shall not contain any long burn time statistical outliers. The results shall be used to set the fuel level for production. (See 6.17)

3.8.2 Delay composition. Each lot of delay composition shall be ballistically tested at 70 degrees F and -50 degrees F. The production batch shall have a mean time of 15.25 $\pm$ .75 seconds and a standard deviation no greater than .62 seconds when ballistically tested at 70 degrees F. For -50 degree F testing, the delays shall function properly to the extent it

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ignites the rocket motor. Static spin stand testing shall be performed to determine suitability for ballistic testing and to establish a baseline for delay assembly testing. During static spin testing, the standard deviation shall not be greater than 0.62 seconds, and no long burn time outliers shall be present.

3.8.3 Delay assembly. Each production lot produced shall have an acceptable static spin mean time per 4.4.8 and 4.4.9 or an acceptable ballistic mean time of 15.25 ± .75 seconds at 70 degrees F and proper function at -50 degrees F when tested ballistically.

3.8.4 Ballistic safety. There shall be no in-bore function (see 6.22 for definitions) of the delay or rocket motor.

3.9 Selector cap. After radiographic inspection of the delay the selector cap shall be assembled hand tight.

3.10 Particle size.

3.10.1 Tungsten particle size. The tungsten shall meet all requirements of MIL-T-48140 for TYPE I (Agglomerated) with a bulk density ranging from 50-65 grams per cubic inch.

3.10.2 Potassium perchlorate particle size. In addition to the requirements of MIL-P-217, the particle size at the 50% level shall be 85 +/- 10 microns.

3.11 Workmanship. The requirements of workmanship are as follows:

a. Cleaning. If required the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

b. Delay assembly. Parts shall be free of burrs, chips, dirt, grease, rust and other foreign matter.

3.11.1 Boron potassium nitrate. The boron potassium nitrate pellets shall not have flashing or any other form of protrusion attached thereto. The boron potassium nitrate charge shall be free of dust, shards, and other types of particles at the time of loading into the delay assembly.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix

Methods of Verification 1 – Analysis 2 – Demonstration 3 – Examination 4 - Test						Classes of Verification A- First Article B- Conformance		
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article			X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2	Components and assemblies			X	X	X	X	4.3.2.1-4.3.2.7, 4.4.2, 4.4.3
3.3	Lot numbering			X	X	X	X	4.3.1
3.4	Pyrotechnic composition, total volatiles			X	X	X	X	4.4.1, 4.3.2.3
3.5	Crimp strength			X	X	X	X	4.4.4, 4.3.2.3
3.6	Pyrotechnic porosity test			X	X	X	X	4.4.5, 4.3.2.3
3.7	Radiographic inspection			X	X	X	X	4.4.10, 4.3.2.5
3.8.1	Pilot composition test			X	X	X	X	4.4.6.1, 4.4.7, 4.4.8
3.8.2	Delay composition test			X	X	X		4.4.6.2, 4.4.7, 4.4.8
3.8.3	Delay assembly testing			X	X	X		4.4.6.3, 4.4.7-4.4.9
3.8.4	Ballistic safety			X	X	X	X	4.3.2.7
3.9	Selector cap			X	X	X	X	4.3.2.6
3.10.1	Tungsten particle size			X	X	X	X	4.4.11, 4.3.2.3
3.10.2	Potassium perchlorate particle size			X	X	X	X	4.4.11, 4.3.2.3
3.11	Workmanship			X	X	X	X	4.3.2.1-4.3.2.6
3.11.1	Boron potassium nitrate			X	X	X	X	4.3.2.4

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4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with Table II.

4.2.1 First article quantity. First article inspection shall be performed on the quantity of items as indicated in Table I.

4.2.2 Inspections to be performed. The first article inspection shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II First article inspection

	Number of FAT Samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Delay assembly (Drawing 12926809) Examination for defects (Note 1) Delay housing crimp test Pilot composition test	 30 5/machine 150	 3.2 3.5 3.8.1	 4.3.2.1-4.3.2.6 4.4.4 4.4.6.1
Note 1. The examination for defects in paragraph 4.3.2 in many cases must be verified during assembly.			

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot numbering shall be in accordance with MIL-STD-1168. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916. In addition, each inspection lot of the M927/M913 Delay Assembly shall contain:

a. Inert parts from one interfix lot number from one manufacturer.

b. Flash composition from not more than one raw material constituent ingredient lot.

c. Delay composition from not more than one inspection lot.

d. Igniter composition from not more than one raw material constituent ingredient lot.

e. Boron Potassium Nitrate pellets from not more than one interfix lot number from one manufacturer.

4.3.1.1 Delay composition lot. A delay composition lot (see 6.22) may be formed by any of the following methods:

a. Blending two or more batches of delay powder together to form a delay composition lot.

b. Designating a batch of delay powder or delay pellets as delay composition lot. Once samples are selected from the batch or delay composition lot, the batch or lot shall not be subsequently blended.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 through 4.3.2.7. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative Inspections. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916. See 6.7.

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PARAGRAPH 4.3.2.1	TITLE Delay assembly, prior to loading delay composition	SHEET 1 OF 1	DRAWING NUMBER 12926809
			NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Flash composition on sidewall	Level IV	3.2	Visual
Minor 201	Evidence of poor workmanship	Level II	3.11	Visual

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PARAGRAPH 4.3.2.2	TITLE Delay assembly, Prior to loading igniter composition	SHEET 1 OF 1	DRAWING NUMBER 12926809 NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 102	Distance from end of housing to delay composition	Level IV	3.2	AIE
Minor 201	Evidence of poor workmanship	Level II	3.11	Visual

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PARAGRAPH 4.3.2.3	TITLE Delay assembly, after crimping washer	SHEET 1 OF 1	DRAWING NUMBER 12926809
			NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total volatiles	4.4.1	3.4	4.4.1
102	Determination of average particle size	4.4.2	3.2	4.4.2
103	Crimp strength	4.4.4	3.5	4.4.4
104	Porosity test	4.4.5	3.6	4.4.5
105	Particle size	4.4.11	3.10	AIE (see 6.11)
106	Washer missing	100%	3.2	Visual
107	Crimp missing or incomplete	100%	3.2	Visual
108	Depth of washer, min.	Level IV	3.2	AIE
109	Determination of VAAR solids content	4.4.3	3.2	4.4.3
Minor				
201	Evidence of poor workmanship	Level II	3.11	Visual

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PARAGRAPH 4.3.2.4	TITLE Delay assembly, prior to assembly of delay closure	SHEET 1 OF 1	DRAWING NUMBER 12926809
			NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Boron potassium nitrate (BKNO3) charge missing	100%	3.2	Visual
102	Weight of BKNO3 charge	Level IV	3.2	Balance/scale
103	Presence of flashing or other protrusions on the boron potassium nitrate pellets prior to loading	Level IV	3.11.1	Visual with Magnification
104	Presence of dust, shards, or other particles in the boron potassium nitrate charge at the time of loading	100%	3.11.1	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.11	Visual

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PARAGRAPH 4.3.2.5	TITLE Delay assembly, prior to assembly of selector cap	SHEET 1 OF 1	DRAWING NUMBER 12926809
			NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Radiographic inspection; delay column missing	Note 1	3.7	4.4.10
Major				
101	Delay closure missing or improperly Positioned.	100%	3.2	Visual
102	Crimp on tube missing or body cracked by crimping operation	100%	3.2	Visual
103	Outside diameter of tube, max.	Level IV	3.2	AIE
104	“O”-ring missing	Level IV	3.2	Visual
105	Gasket missing in cap	100%	3.2	Visual/AIE
106	Radiographic inspection; defects other than ‘missing delay column’	100%	3.7	4.4.10
Minor				
201	Evidence of poor workmanship	Level II	3.11	Visual

Note 1. 100% inspection by Level II or better certified reader with a 1 in 10 review by a Level III certified non destructive test inspector.

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PARAGRAPH 4.3.2.6	TITLE Delay assembly		SHEET 1 OF 1	DRAWING NUMBER 12926809 NEXT HIGHER ASSEMBLY -
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Selector cap missing or not hand tight	Level IV	3.9	Visual/manual
102	Identification marking missing or incorrect	Level IV	3.2	Visual
103	Rocket selector cap O-ring or gasket missing	Level IV	3.2	Visual
104	Lubricant not applied to O-ring	Level IV	3.2	Visual
105	Lubricant on face of rocket cap assembly	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.11	Visual

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PARAGRAPH 4.3.2.7	TITLE Delay assembly, performance tests	SHEET 1 OF 1	DRAWING NUMBER 12926809
			NEXT HIGHER ASSEMBLY -

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Pilot composition test, in-bore delay function	4.4.6.1	3.8.1/3.8.4	4.4.6.1
2	Delay composition lot test, in-bore delay function	4.4.6.2	3.8.2/3.8.4	4.4.6.2
3	Delay assembly test, in-bore delay function	4.4.6.3	3.8.3/3.8.4	4.4.6.3
Major				
101	Pilot composition test, no delay function or burn time incorrect	4.4.6.1	3.8.1	4.4.6.1
102	Delay composition lot test, no delay function or burn time incorrect	4.4.6.2	3.8.2	4.4.6.2
103	Delay assembly test, no delay function or burn time incorrect	4.4.6.3	3.8.3	4.4.6.3
104	Pilot composition test, static burn time	4.4.6.1	3.8.1	4.4.6.1
105	Delay composition lot test, static burn time	4.4.6.2	3.8.2	4.4.6.2
106	Delay assembly test, static burn time	4.4.6.3	3.8.3	4.4.6.3

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4.4 Testing.

4.4.1 Total volatiles. For verification, a sample of delay, igniter and flash composition from each regular production shift or fraction thereof and from each new container shall be selected to perform a total volatiles analysis. If any sample fails to meet the requirements of 3.4, the production represented by the sample (or the new container of material) shall be rejected. The sample shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.4.2 Determination of average particle size.

4.4.2.1 Molybdenum trioxide. A sample shall be selected from each lot of molybdenum trioxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 12926809 for particle size, the lot shall be rejected.

4.4.2.2 Chromium oxide. A sample shall be selected from each lot of chromium oxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 12926809 for particle size, the lot shall be rejected.

4.4.3 Vinyl Alcohol Acetate Resin (VAAR) solids determination. The concentration of solids in VAAR solution may change during storage due to evaporation of solvent. VAAR solids content in the delay blend can only be obtained if the solution to solids weight ratio is known prior to mixing. Prior to blending the delay composition, a sample of VAAR solution shall be tested to determine this ratio. The sample shall be tested in accordance with and meet the requirements of MIL-V-50433. Should the solids content be too high, additional solvent may be added, with agitation, until the solids content is acceptable. The VAAR solution shall then be retested to assure compliance with MIL-V-50433.

4.4.4 Crimp strength. At the beginning of each production shift or portion thereof, an inert delay housing shall be crimped on each crimping device using the same technique as in regular production. After crimping, the delay housing shall be sectioned perpendicular to the longitudinal axis and the load applied axially to crimp from the sectioned end. A tool manufactured to the same diameter as the delay column bore, minimum, shall be used in applying load to the crimp. If the crimp fails to meet the load requirements of 3.5, the crimping device shall be adjusted until the requirements are met. (See 6.10)

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4.4.5 Porosity test. This test shall be performed 100% until 1500 consecutive units have passed without failure. After the above criteria has been met, sampling shall be performed randomly at the rate of 18 delay assemblies for each production shift or portion thereof. If any one of the 18 delay assemblies fail to meet the requirements of 3.6, the total shift production shall undergo the porosity test and an additional 1500 units tested to quality sampling again. (See 6.13) Each delay assembly shall be conditioned at 145 +/- 5 degrees F and assembled to the porosity test equipment. Nitrogen gas under a pressure of 1000 PSIG plus or minus 50 PSIG shall be applied to the delay column at the threaded end of the housing. The pressure at the exit end of the delay column shall be recorded at 0.5 seconds, 80 seconds and 110 seconds. The pressure reading at 0.5 seconds shall be recorded and used as the zero reference.

4.4.6 Performance testing.

4.4.6.1 Pilot composition test. For first article and each new lot of constituent ingredients, a pilot composition test shall be performed on 150 delay assemblies utilizing both ballistic and static spin testing. Five pilot batches are to be produced at tungsten contents differing from sample to sample by a constant difference. The difference shall not exceed one part by weight. From each of the five completed batches of delay composition, enough material shall be selected to load 30 delay assemblies. If the sample is taken before the entire batch is pelletized, the representative sample shall be selected in 30 approximately equal increments taken from different areas in the mix, tumbling the mix often so as to minimize the chance of misleading results from segregation. If the sample is taken from a pelletized batch, a random sample of pellets shall be selected for use in this test. All thirty delays from each batch shall be loaded with igniter, delay composition, flash, and have the washer installed. Fifteen of the delays shall then be selected at random for static spin testing and the remaining delays shall be loaded with the boron potassium nitrate and delay closure for ballistic testing. All delays shall have the selector caps installed. Selector caps used for spin testing may be reused only for other spin test delays provided care is taken to assure pyrotechnic material is removed from the cap and gasket prior to reuse. All thirty delays from each lot shall be porosity tested and radiographically inspected per 4.4.5 and 4.4.10 respectively and must meet the requirements of 3.6 and 3.7 respectively. Fifteen delay assemblies shall be statically spin tested in accordance with 4.4.8, and the results tested for outliers per 6.12. If the spin test results for the batch exhibit no long time outliers per 6.12, the remaining fifteen completely loaded delay assemblies shall be ballistically tested in accordance with 4.4.7. If the requirements of 3.8.1 are not met, or if the

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delay fails to function, the batch shall be considered to have a Major defect and shall not be used for future production. If, during ballistic testing, the delay ignites the motor grain while in the gun tube, the batch shall be considered to have a Critical defect, and the testing shall cease. A minimum of 8 radar delay times shall be required to determine delay time acceptance for each batch during the 70 degree F temperature phase. All delays shall function properly when ballistically tested at -50 degrees F.

4.4.6.2 Delay composition lot test. From each production lot of delay composition, a random sample shall be selected and loaded into 30 sample delay assemblies. Fifteen sample delays shall be statically spin tested to determine mean time, standard deviation and the presence of long time outliers per 6.12. If the standard deviation requirement is met and no outliers are detected, 15 complete delay assemblies shall be produced and forwarded for ballistic testing per 4.4.7. If the sample fails to meet the requirements of 3.8.2 it is considered a Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance.

4.4.6.3 Delay assembly test. From each lot of delay assemblies, 15 samples shall be randomly selected and statically spin tested. Using the spin test baseline data generated from 4.4.6.2, a determination of acceptability shall be made per 4.4.9. If the spin test results are not acceptable, then fifteen completed delay assemblies shall be selected at random for ballistic testing per 4.4.7. If the sample fails to meet the requirements of 3.7.3 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance.

4.4.6.4 Additional test for outliers. If the spin test data meets the requirement for standard deviation, but one or more outliers are detected, the 15 remaining samples may be spin tested. The mean time, standard deviation and presence of long time outliers (see 6.12.1) shall be calculated for the sample of 30. If the standard deviation requirement is met and no outliers are detected, 15 additional complete delay assemblies shall be produced and forwarded for ballistic testing per 4.4.7.

4.4.7 Ballistic test. The delay assemblies called out for ballistic testing per 4.4.6 shall be forwarded to a facility (see 6.2) for loading into 105MM M913 or M927 cartridges for ballistic testing at a proving ground. To assure a proper seal the loaded delay assemblies shall be radiographed after pressing into the

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motor body. A fuze and selector cap shall be assembled to the test vehicle prior to temperature conditioning. Ten rounds, including propelling charges, shall be conditioned at 70 +/- 2.5 degrees F for a minimum of 24 hours. Five rounds shall be conditioned to -50 -5 degrees F for a minimum of 24 hours. The selector cap shall be removed and the round fired from an M119 series howitzer with a minimum of 25 percent tube life remaining. An M229 or M67 propelling charge shall be used in these tests. At least two smear cameras shall be mounted near the gun muzzle to photograph the projectile in flight. One camera shall provide a side view to assure metal parts integrity. Another camera shall provide a rear view to determine presence and ignition of the delay assembly. Delay time shall be determined by use of radar. Stopwatch times may be used as a back-up to confirm that the delay functioned but shall not be used in computations for delay acceptance.

4.4.8 Static spin stand testing. The fifteen delay assemblies that shall be furnished for this test shall be comprised of delay housing with the flash, igniter, delay composition, washer, and selector cap assembled. The delay assemblies shall be porosity tested and radiographically examined prior to spin testing. The apparatus for spin stand testing shall be comprised of a stand suitable for safe ignition of the delay column that has a fixture that shall hold the delay assembly, without slippage, while it is rotating along its longitudinal axis at 20,000 RPM. An electronic squib fixtured near the washer opening shall be used to initiate the igniter after the fixtured delay has stabilized at 20,000 +/-200 RPM for the M913 or 15,000 +/-200 RPM for the M927. An electronic timer triggered by the squib ignition and terminated by the flash charge output shall display total burn time. A tachometer shall be used to verify delay spin. The temperature of the delay containment fixture and delay shall be 75 +/- 15 degrees F before performing each spin test. Delay burn time shall be observed to the nearest 0.1 second and the average delay burn time and standard deviation shall be calculated for each batch tested. This data shall be used for determination of long time outlier (see 6.12).

4.4.9 Comparison test for production batch acceptance. The difference between ballistic flight and static spin mean times from the delay composition test shall be added to the static spin mean time of the lot being tested. If the calculated mean time falls within the requirement of 3.8, the lot shall be accepted.

4.4.10 Radiographic test for proper loading. All delays shall be subjected to radiographic examination in accordance with

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drawing 13023959. Examination of the radiograph shall be made to determine compliance with 3.7. (See 6.4.1)

4.4.11 Particle size. Prior to mixing of each composition batch, or pilot mixture, a sample of tungsten and of potassium perchlorate shall be tested for particle size from each shipping or storage container from which chemicals shall be used in the mix. If the particle size of any sample tested fails to comply with the requirements of 3.10, chemical from that container shall not be used in the mix. Chemicals may be retested for use after application of suitable rework. A sample of 15g minimum shall be taken from each individual container in such a manner that the sample is representative of the entire contents of the container. No lab milling, mulling, grinding or similar operation is permitted during sampling or preparation thereof. For test method guidance see 6.11.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department or Defense Agency, automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The delay assemblies covered by this specification are intended for use on the 105MM, HE, RA, Projectiles M913 and M927. This specification contains requirements suitable for production.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Requirements for submission of first article sample.

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c. Submission of contractor acceptance inspection equipment. (See 6.4)

d. Provisions for Certification of nondestructive test (NDT) personnel. (See 6.3)

e. Submission of ballistic test data and ammunition data cards. (See 6.4)

f. Provisions for critical characteristic controls. (See 6.6)

g. Location where proving ground tests should be conducted.

6.3 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment should be qualified and certified in accordance with NAS-410 “Certification and Qualification of Nondestructive Test Personnel”. Application for copies should be addressed to the Aerospace Industries Association of America, Inc, (AIA/NAS), 1000 Wilson Blvd, Suite 1700 Arlington, VA 22209. Qualification to other standards must be submitted to and approved by Commander, ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000.

6.4 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-A, Picatinny, NJ 07806-5000. This address should be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4.1 Radiographic procedures. Radiographic procedures and sample radiographs should be submitted to the Commander, ARDEC, Picatinny Arsenal, 07806-5000, ATTN: RDAR-QEM-A for review and approval upon implementation at each six month period of production.

6.5 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards should be forwarded to Commander, ARDEC, ATTN: RDAR-AEM-C and RDAR-QEM-A, Picatinny, NJ 07806-5000.

6.6 Critical characteristic controls. All production contracts should contain requirements for handling critical defects. As part of all proposals, suppliers should be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier’s integrated management system.

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6.7 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions may be submitted to the Government for evaluation/approval.

6.8 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.9 Consolidation load. The contractor should provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. Proper calibration of each consolidation machine should be verified at the beginning, middle and end of each production shift. Dwell time and consolidation load should be measured with a stopwatch and a load cell. If any consolidation machine fails verification for dwell time, consolidation load or for not maintaining proper load through the dwell time, all the delays manufactured on that machine since the last acceptable verification should be rejected. In addition, a warning device indicating failure of dwell time and failure of consolidation load should be assembled to each consolidation machine. The warning device or mechanism should provide an easily detected signal to the machine operator. If during loading of the delay the consolidation load or dwell time requirements are not met, that delay should be rejected.

6.9.1 Process control for delay increment consolidation. A process control description should be developed for delay increment consolidation, and submitted for Government approval. The description should include all elements of press setup, operation, calibration, maintenance and operator instruction required to assure that the required number of delay increments are individually consolidated at the required pressure and for the required minimum consolidation time. The process controls should include verification that the required pressure and dwell are met at the extremes of the press consolidation range (See 4.4.3.1 and 6.9).

6.10 Crimping Process. The crimping process should be monitored to assure that proper crimp strength is maintained throughout production.

6.11 Particle size.

6.11.1 Particle size analyzer. Sedigraph particle analyzer 5000D or an approved equivalent instrument that uses a finely collimated beam to measure particle concentration as a function of time is recommended.

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6.11.1.1 Dispersing fluid. The sedimentation fluid should be Sedisperse A12, a product of Micromeritics, or an approved equivalent.

6.11.2 Sedigraph test sample. Fill a 25 ml beaker with dispersing fluid and adequate sample powder for 100% recorder pen deflection when the beaker is placed in test position and the instrument is running and has been adjusted to section 7-12 of ASTM-B430.

6.11.2.1 Carrier density/viscosity. If sedisperse A12 is used as a carrier, the carrier density and viscosity is supplied on the bottle label by Micromeritics for one degree temperature changes in a range of 24-36 degrees centigrade. If another carrier is used, then the user must establish these temperature dependent physical properties, before measurements.

6.11.2.2 Sedimentation rates. Sedimentation rates may be calculated using procedure outlined in MIL-M-382 under 4.5.11.2 with these exceptions:

a. P = density of sedisperse A12 or another compatible fluid.

b. n = viscosity of sedisperse A12 or another compatible fluid.

6.12 Dixon test for statistical outliers. This test is to be performed on static spin delay times only. For determining a long-time outlier at the 95% level of confidence, order the delay burn times from longest to shortest (T1-T15 with T1 being the longest). Calculate the Dixon ratio (R) by the following:

$$R = \frac{(T3-T1)}{(T13-T1)}$$

(Note that T15 is the shortest time)
If R is greater than .525, then the longest time is an outlier.

6.12.1 Dixon test sample size of 30. Order the delay burn times from longest to shortest (T1 – T30 with T1 being the longest). Calculate the Dixon ration (R) by the following:

$$R = \frac{(T3 - T1)}{(T28 - T1)}$$

If R is greater than .376, then the longest time is an outlier.

6.13 Porosity test equipment. The contractor should design and construct a porosity test fixture capable of testing delays

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to the parameters 4.4.5. M549 drawings 9246401 and 9246420 may be used as a guide.

6.14 Handling of delay assemblies. Precaution should be taken during loading, intraplant shipment for in process test, subsequent to loading and prior to packing to assure delays are not exposed to moisture. Delay assemblies should not be removed from moisture proof packing until such time as the units are to be placed in the conditioning chamber for porosity testing and flight testing.

6.15 Source of supply. The model 5000D Sedigraph can be secured from Micromeritics Inst. Corp.; Norcross, Georgia, USA.

6.16 Submission of process control description. The process control description for delay increment consolidation should be prepared in accordance with DIP-1604 and submitted to the responsible Government technical agency for approval.

6.17 Static spin versus ballistic delay burn times. Testing has shown that delay assemblies burn approximately 0.75 to 1.25 seconds faster on the static spin stand than in ballistic testing.

6.18 Fuel ratio. The exact fuel/oxidizer ratios for the pilot batches should be determined by the contractor.

6.19 Critical classification notes.

6.19.1 Radiographic inspection; delay column missing. A missing delay column is likely to cause an in-bore delay function, which can cause a rocket motor separation (from the warhead) upon muzzle exit. This is likely to result in the warhead landing critically short which may lead to unintended casualties.

6.19.2 In-bore delay function. An in-bore delay function can cause a rocket motor separation (from the warhead) upon muzzle exit. This is likely to result in the warhead landing critically short which may lead to unintended casualties.

6.20 Ballistic test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test should be considered invalid and a new test should be performed with additional samples.

6.21 Material rejected for total volatiles. Drying of rejected material should be accomplished with specific instructions from the Government procuring activity.

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6.22 Definitions.

6.22.1 Delay composition lot. A delay composition lot is defined as that quantity of delay material from which representative samples have been selected, loaded and forwarded to a proving ground for acceptance.

6.22.2 Pilot composition test. The pilot composition test phase is performed to determine the correct amount of tungsten fuel to be used to achieve the requirements as well as tests for mean time, standard deviation, and outliers using static spin stand testing and ballistic testing.

6.22.3 Delay composition test. The delay composition test ballistically qualifies the production batch prior to production assembly and establishes a baseline for spin stand time testing, also checking for outliers.

6.22.4 Delay assembly test. The delay assembly test for mean time and standard deviation is performed on a sample of the loaded inspection lot, using static spin stand testing and ballistic testing as required.

6.22.5 In-bore delay function. An in-bore delay function is defined as the flash of the boron potassium nitrate pellets in a way that ignites or is likely to ignite the rocket motor in the gun tube. In-bore delay function is evinced by photographic or radar indication that the rocket motor was burning at the weapon muzzle, that the projectile emerged from the muzzle in two or more pieces, or that the delay discarded at or immediately after muzzle exit.

6.23 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.24 Subject term (key word) listing.
Pyrotechnic delay

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1315-2009-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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